

2(a)(i)	(vertically) downwards	B1
2(a)(ii)	magnetic force (on sphere) is perpendicular to its velocity	B1
	magnetic force perpendicular to velocity is the centripetal force or magnetic force perpendicular to velocity causes centripetal acceleration or acceleration perpendicular to velocity is centripetal (acceleration) or magnetic force does not change the speed of the sphere or magnetic force has constant magnitude	B1
2(b)	$mg = Eq$	C1
	$E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8 \text{ N C}^{-1}$	A1
2(c)	centripetal force = magnetic force or $Bqv = mv^2 / r$	B1
	$B = mv / qr$	C1
	$= (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14 \text{ T}$	A1